

Chapter 18

Point charges and the energy–momentum tensor

We now turn to electrodynamics – charges in motion.

18.1 Point charges

A point charge q moving along a trajectory $\vec{y}(t)$ carries the current density

$$\vec{J} = q \vec{v} \delta^3(\vec{x} - \vec{y}) \quad (18.1)$$

This is packaged covariantly into the four-current

$$j^\mu(x) = q c \int \frac{dy^\mu}{d\tau} \delta^4(x - y(\tau)) d\tau \quad (18.2)$$

where $y^\mu(\tau)$ is the worldline of the charge. Is this conserved, $\partial_\mu j^\mu \stackrel{?}{=} 0$? Treat j^μ as a distribution and smear against an arbitrary bounded function $f(x)$:

$$\begin{aligned} \partial_\mu j^\mu[f] &= \int \partial_\mu j^\mu f(x) d^4x = - \int j^\mu \partial_\mu f d^4x \\ &= - \int \int q c \frac{dy^\mu}{d\tau} \delta^4(x - y(\tau)) \partial_\mu f d^4x d\tau \\ &= -q c \int \frac{dy^\mu}{d\tau} \partial_\mu f(y) d\tau = -q c \int df(y) \\ &= -q c (f(y(\infty)) - f(y(-\infty))) = 0 \quad (18.3) \end{aligned}$$

since $y(\pm\infty) \rightarrow \infty$. Hence $\partial_\mu j^\mu = 0$.

To check the components of (18.2), trade the proper-time integral for a coordinate-time integral

$$\begin{aligned} j^\mu(x) &= q c \int \frac{dy^\mu}{d\tau} \delta^4(x - y) d\tau = q c \int \frac{dy^\mu}{dt} \delta^3(\vec{x} - \vec{y}) \delta(ct - y^0) dt \\ &= q \frac{dy^\mu}{dt} \delta^3(\vec{x} - \vec{y}) \Big|_{y^0=ct} \end{aligned} \quad (18.4)$$

so that

$$j^0 = q c \delta^3(\vec{x} - \vec{y}(t)) \quad , \quad \vec{j} = q \vec{v}(t) \delta^3(\vec{x} - \vec{y}(t)) \quad (18.5)$$

as expected.

18.2 Energy and momentum

We learned that the energy density of the electromagnetic field is

$$u = \frac{1}{8\pi} (E^2 + B^2) \quad (18.6)$$

We must now write this in Lorentz covariant form (as we did for energy with the four-momentum). Generally, any conserved quantity – “garbage” – comes with a four-vector current j_G^μ obeying the conservation statement $\partial_\mu j_G^\mu = 0$, with

$$\frac{1}{c} j_G^0 = \text{garbage density} \quad , \quad j_G^i = \text{garbage flux} = \frac{\text{garbage}}{\text{time} \cdot \text{area}} \quad (18.7)$$

For $\vec{E}$ and $\vec{B}$ we need an energy density with its energy flux, and a momentum density with its momentum flux. Hence we need a $(2, 0)$ tensor $T_{\text{EM}}^{\mu\nu}$ – four currents at once

$$j_{(\mu)}^\nu \equiv T_{\text{EM}}^{\mu\nu} \quad , \quad \mu = 0, 1, 2, 3 \quad (18.8)$$

with $\mu = 0$ the current of time translation symmetry, and $\mu = i$ the currents of space translation symmetry. We expect

$$T_{\text{EM}}^{00} = u c \quad , \quad T_{\text{EM}}^{0i} \equiv S^i = \text{energy flux} \quad (18.9)$$

with u the energy density, and

$$\frac{1}{c} T_{\text{EM}}^{i0} = (\text{momentum density in the } i\text{th direction}) \times c \quad (18.10)$$

and

$$T_{\text{EM}}^{ij} = \text{momentum flux} \times c = \text{stress} \times c \quad (18.11)$$

To find $T_{\text{EM}}^{\mu\nu}$, write the most general combination quadratic in the field strength

$$T_{\text{EM}}^{\mu\nu} = c_1 F^\mu{}_\alpha F^{\alpha\nu} + c_2 \eta^{\mu\nu} F^{\alpha\beta} F_{\alpha\beta} + c_3 F^{\mu\nu} F^\alpha{}_\alpha \quad (18.12)$$

where the last term vanishes since $F^\alpha{}_\alpha = 0$ by antisymmetry. Computing the 00 component,

$$T_{\text{EM}}^{00} = -(c_1 - 2c_2) E^2 - 2c_2 B^2 = u c \quad (18.13)$$

which fixes c_1 and c_2 , giving

$$T_{\text{EM}}^{\mu\nu} = -\frac{c}{4\pi} \left(F^\mu{}_\alpha F^{\alpha\nu} + \frac{1}{4} \eta^{\mu\nu} F^{\alpha\beta} F_{\alpha\beta} \right) \quad (18.14)$$

Note that $T^{\mu\nu} = T^{\nu\mu}$ – a symmetric tensor. The remaining components are then

$$T_{\text{EM}}^{0i} = \frac{c}{4\pi} \left(\vec{E} \times \vec{B} \right)^i = S^i \quad (18.15)$$

the Poynting vector, which is also c^2 times the momentum density, and

$$T_{\text{EM}}^{ij} = \frac{c}{4\pi} \left[-\frac{1}{2} (E^2 + B^2) \delta^{ij} + E^i E^j + B^i B^j \right] \quad (18.16)$$

which is the stress times c .

The total energy–momentum consists of a part from the electromagnetic field plus a part from the sources. For a point particle, we must have

$$T_{\text{particle}}^{\mu\nu} \propto u^\mu u^\nu \quad (18.17)$$

(by Lorentz symmetry plus the non-relativistic limit), that is

$$\begin{aligned} T_{\text{particle}}^{\mu\nu} &= m \int u^\mu u^\nu \delta^4(x - y(\tau)) d\tau = m \int u^\mu \frac{dy^\nu}{dt} \delta^4(x - y) dt \\ &= m \int u^\mu \delta^3(\vec{x} - \vec{y}) \delta(x^0 - y^0) dy^\nu \end{aligned} \quad (18.18)$$

Check the components:

$$\begin{aligned} T_{\text{particle}}^{00} &= m c \int u^0 \delta^3(\vec{x} - \vec{y}) \delta(x^0 - y^0) dt = m u^0 \delta^3(\vec{x} - \vec{y}) \Big|_{y^0=ct} \\ &= \gamma m c \delta^3(\vec{x} - \vec{y}) \end{aligned} \quad (18.19)$$

as expected, and

$$T_{\text{particle}}^{i0} = m u^i \delta^3(\vec{x} - \vec{y}) = \gamma m v^i \delta^3(\vec{x} - \vec{y}) \quad (18.20)$$

as expected. For the particle plus field system,

$$T^{\mu\nu} = T_{\text{EM}}^{\mu\nu} + T_{\text{particle}}^{\mu\nu} \Rightarrow \partial_\mu T^{\mu\nu} = 0 \quad (18.21)$$

Note also that the electromagnetic energy–momentum tensor is traceless

$$T_{\text{EM}}^{\mu\nu} \eta_{\mu\nu} = 0 \quad (18.22)$$

(a reflection of scale invariance).